QND Computer Science Day 2

Mr. Schmidt

Recap

- Website
- Swift Playgrounds
- Strings
- `console.write`
- Variables

Agenda

- Text Console
- `console.write`
- `console.read`
- Concatenation
- If Statements?

Hello World!

Type the following into the code editor:

```
console.write("Hello World!")
```

Then tap the ▶️ **Run My Code** button on the bottom right

You should see `Hello World!` appear in the console!

Strings

- A sequence of characters
- Characters
  - Letters, numbers, punctuation, emoji 🚀🚀🚀
- Always between quotation marks `" "`

The `write` Function

- `console.write("Hello 🦀🦀🦀")`
- Writes the provided string to the console output
- Add emoji with `control` + `command` + `spacebar`
- We will see many, many other functions
- `function(input1, input2, input3...)`

Variables

- *Declare* variables with `let`
- Just like in math, replaces the variable with a value
- Read it as "Let greeting equal ..."

```
let greeting = "Hello 🦀🦀🦀!"
console.write(greeting)
console.write(greeting)
console.write(greeting)
```

Concatenation

- Combine variables together!

```
let greeting = "Hello "
let name = "Mr. Schmidt"
console.write(greeting + name)
```

This program is boring!

- It needs to respond to user input
- Use `console.read`
- Waits for the user to type in the console and press Enter
- Stores the value in a variable

```
let name = console.read("What is your name?")
console.write("Hello " + name)
```

- Use concatenation to say hello!

What is the Song of the Summer?

- A certified bop
- A jam and a half

Qualities of the Song of the Summer

- Beach vibes
- Upbeat, danceable
- Simple chorus -- you only need to hear it once!

Possible Topics for the Song of the Summer

- My ex is awful (Good 4 U)
- My SO is great (Cheerleader)
- Dancing is fun (Hot to Go)
- My haters are trash (Not Like Us)

What is NOT the Song of the Summer?

- Whatever is #1 on the charts
- Sad girl autumn crash out music
- Whatever is big on TikTok
  - The people choose the song of the summer, not The Algorithm

`if`

- Execute different code based on a condition
- Brackets define what code is executed if the condition is true

```
let answer = console.read("What's the song of the summer?")
if answer == "Drop Dead" {
    console.write("Correct!")
} else {
    console.write("Wrong!")
}
```

`else if`

- `if` statements can be chained together with `else if`
- The first condition that is true will be executed
- Add any number of else if statements **in between** `if` and `else`

```
let answer = console.read("What's the song of the summer?")
if answer == "Drop Dead" {
    console.write("Correct!")
} else if answer == "Choosin Texas" {
    console.write("Hmm, not quite!")
} else {
    console.write("Wrong!")
}
```

OR

- `==` is case sensitive and spelling sensitive
- How can we make sure that both `"Drop Dead"` and `"drop dead"` are allowed?

```
if answer == "Drop Dead" || answer == "drop dead" {
    ...
}
```

- `||` is the OR operator
- Note: `answer == "Drop Dead" || "drop dead"` is not correct -- the `||` must go between two full comparisons

Assignment

- Add more branches!